

Human Group Strategies: An Agent-Based Simulation of Cooperation, Retaliation, and Expansion

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Abstract

This work studies how the strategy of a human group affects its survival and prosperity in a spatial world with limited resources. We implement a stochastic agent-based model in which each country controls territorial grid cells, grows toward a territory-dependent carrying capacity, explores free land, fights neighbouring countries, and forms trade, non-aggression, and military pacts. Five fixed strategies are compared: peaceful cooperation, armed neutrality, aggressive hegemony, retaliation, and a random baseline. The analysis comprises 2,500 simulation runs: 100 seeds for ten scenarios and three five-level parameter sweeps. No strategy is uniformly optimal. Hegemony has the strongest baseline performance (85% survival and 39% dominance), but retaliation becomes the most frequently dominant strategy on a spacious map (49%) and under direct deterrence (31%). Stronger defence substantially improves peaceful and neutral strategies, whereas a random strategy nearly disappears. The results show that success depends jointly on resource pressure, combat asymmetry, population recovery, and—most importantly—the strategies present in the surrounding population.

1 Introduction

Human groups compete for land and resources, but they also trade, build alliances, merge, and sometimes avoid conflict entirely. These interactions form a feedback system: territorial expansion increases carrying capacity; population affects military strength; war changes relationships; and relationships determine whether future cooperation is possible. Even when the rules used by each group are simple, the resulting spatial and diplomatic patterns can be difficult to predict by inspection.

Agent-based modelling is appropriate for this type of question because it represents autonomous decision makers and their local interactions directly. It is especially useful when collective behaviour emerges from nonlinear rules, thresholds, and stochastic events [1]. Classic spatial models demonstrate that simple local preferences can create strong aggregate patterns [2], while artificial-society models show how trade, combat, and resource constraints can jointly generate complex histories [3]. Reciprocity is also a well-established route by which cooperation can persist among self-interested actors [4].

The objective of this project is therefore not to predict a real society. It is to construct a transparent computational experiment answering the question: *under which environmental and strategic conditions does a group strategy lead to survival and prosperity?* Prosperity is evaluated by final population, controlled territory, and whether a country is the largest surviving country at the end of a run. Survival time and behavioural measures such as attacks, casualties, and pacts are recorded to explain those outcomes.

2 Model

The model is a stochastic, discrete-time, spatial agent-based simulation. A country is the unit of agency. It has a population, territory, partial knowledge of the world, bilateral relationships, and one fixed strategy. A run stops after 300 turns or when at most one country remains.

2.1 World

The world is a rectangular grid with a von Neumann neighbourhood. Each cell is either unclaimed or controlled by one country. All cells have equal resource productivity: one cell supports 1,000 people. A country may explore only an unclaimed cell sharing an edge with its territory, and it may attack only an enemy cell sharing such an edge. New territory is therefore acquired locally rather than teleported across the map.

For country i , let $A_i(t)$ be its number of cells, $P_i(t)$ its population, and $T_i(t)$ its number of active trade partners. Its carrying capacity is

$$K_i(t) = 1000A_i(t) [1 + 0.1 \min\{T_i(t), 5\}]. \quad (1)$$

Trade is the model's representation of food sharing: each partner increases capacity by 10%, up to five partners. Population follows logistic growth with rate r while it is below capacity. If a loss of territory leaves the population above capacity, 20% of the current population is lost per turn until capacity is reached:

$$P_i(t+1) = \begin{cases} \min\{K_i, P_i + rP_i(1 - P_i/K_i)\}, & P_i \leq K_i, \\ \max\{K_i, 0.8P_i\}, & P_i > K_i. \end{cases} \quad (2)$$

Unless varied experimentally, the world is 16×16 , $r = 0.05$, every country starts with 100 people and one

cell, and starting cells are scattered as far apart as the grid allows.

2.2 Group

A country stores a single aggregate population rather than individual people. Its state consists of population, owned cells, known cells, strategy, recent attackers, and one directional attitude toward every other country. Two countries establish contact only after their territories share a border. They then exchange their known maps, although diplomacy with a third country still requires direct contact.

Each living country takes at most one action per turn. The available actions are: explore a free neighbouring cell, attack a neighbouring enemy cell, offer one pact component, or propose a union. The order of countries is reshuffled every turn to avoid a permanent first-mover advantage. A proposed action is checked again immediately before execution because an earlier country may already have changed the world.

2.3 Inter-group interactions

2.3.1 Diplomacy and pacts

An attitude $R_{ij} \in [-100, 100]$ describes how country i views country j ; it need not equal R_{ji} . In the absence of new events, attitudes drift one point per turn toward zero. An attack reduces the defender’s attitude by 50, losing a cell reduces it by another 10, and intervention by an enemy ally reduces it by 20.

Pacts are symmetric but require a minimum mutual attitude. Trade requires 0, non-aggression requires 30, and a military alliance requires 50. Active pacts add 2, 3, or 5 attitude points per turn, respectively. A pact lapses when mutual attitude falls more than ten points below its threshold. An attack breaks non-aggression and alliance commitments and damages the attacker’s reputation with its other pact partners. Alliances are not transitive: a country allied with both combatants remains neutral, and only an ally sharing a border with the opponent can contribute military power.

A union is possible only between neighbours with a military alliance and mutual attitude of at least 80. If both strategies accept, the larger population absorbs the smaller country, including its people, territory, knowledge, and diplomatic connections.

2.3.2 War

Military power balances total population against the difficulty of projecting force over a large territory:

$$S_i(t) = \frac{P_i(t)}{\sqrt{A_i(t)}}. \quad (3)$$

An eligible ally contributes half of its own S_i . If S_A and S_D are the resulting attacker and defender strengths and d is the defence multiplier, the probability of an attacker victory is

$$\Pr(A \text{ wins}) = \frac{S_A}{S_A + dS_D}. \quad (4)$$

The default is $d = 1.5$. Combat concerns one border cell. Its population is approximated by the defender’s population divided by its territory. If the attacker wins, it takes the cell, assimilates half of those inhabitants, and loses 20% of the cell’s population in casualties. If it loses, its side loses 40%. Allies share losses in proportion to contributed power. A country ceases to exist after losing its final cell, population collapse, or absorption by union.

2.4 Strategies

Strategies combine three action probabilities with categorical rules about acceptable targets and agreements. Table 1 summarizes the five implementations. The random strategy is a deliberately weak baseline: it chooses uniformly from all currently legal actions.

The ordinary attack rule rejects a target below a 30–35% estimated win chance, depending on strategy. Negative attitude lowers this threshold, so hostility can sustain conflict. Retaliation instead uses a ten-turn memory and ignores the estimated odds, making the threat credible but sometimes costly.

3 Experimental Design

3.1 Scenarios and parameter sweeps

Ten scenarios test environmental pressure and strategic composition (Table 2). The baseline contains one country of every strategy. Five environmental variants retain that composition while changing map width, defence, or growth. Four context scenarios change the mix of strategies to isolate the effect of neighbours.

In addition, three one-factor sweeps use the baseline composition. Map width takes values 8, 12, 16, 20, 26; the defender multiplier takes 1.0, 1.5, 2.0, 2.5, 3.0; and growth rate takes 0.02, 0.035, 0.05, 0.075, 0.10. Each level is repeated for 100 seeds. In total, $10 \times 100 + 3 \times 5 \times 100 = 2,500$ stochastic runs produce 5,100 country observations for named scenarios and 7,500 observations for sweeps.

3.2 Measures and reproducibility

The primary measures are survival at the final turn, survival time, final population, final territory, territory share, and dominance. A country is “dominant” when it is the largest surviving country by territory, with population breaking ties. Secondary measures include attacks, casualties, cells gained and lost, pacts, and time in alliances.

All comparative plots use repeated runs. Lines in parameter sweeps show arithmetic means and 95% Monte Carlo intervals computed as $\bar{x} \pm 1.96s/\sqrt{n}$. The final-map and dynamics figures use seed 0 only and are presented as illustrations, not estimates. Raw run-level CSV files, aggregated CSV files, representative time series, final maps, and a JSON manifest are written before plotting. Consequently, figures can be regenerated without rerunning the simulation:

```
uv run analysis.py collect --runs 100 \
```

Table 1: Implemented strategy rules and trait probabilities. Diplomacy is the propensity to offer or accept pacts; aggression is the probability of using an acceptable attack; union is the propensity to consolidate.

Strategy	Diplomacy	Aggression	Union	Defining rule
PEACE	0.90	0.00	0.50	Never attacks and may accept every pact type.
SWITZERLAND	0.90	0.00	0.00	Never attacks and refuses military alliances and unions.
HEGEMONY	0.20	0.95	0.00	Attacks any viable neighbour, refuses peace commitments, but may trade.
RETALIATION	0.70	0.95	0.30	Attacks only a country that attacked it in the previous ten turns, regardless of the odds.
RANDOM	0.50	0.50	0.50	Selects uniformly from all legal actions.

Table 2: Scenario design. All unlisted parameters retain their baseline values. Each scenario was repeated for seeds 0–99.

Scenario	Map / parameter	Composition	Purpose
Baseline	16×16	one of each strategy	General comparison
Crowded	8×8	one of each	Immediate land scarcity
Spacious	26×26	one of each	Weak resource pressure
Strong defence	$d = 3.0$	one of each	High cost of conquest
Weak defence	$d = 1.0$	one of each	No defender advantage
Slow growth	$r = 0.02$	one of each	Slow recovery after war
One aggressor	16×16	1 hegemony, 4 peace	Aggressor among cooperators
Many aggressors	16×16	4 hegemony, 1 peace	Competition among aggressors
Deterrence	16×16	2 hegemony, 2 retaliation, 1 peace	Credible response to aggression
Neutrals	18×18	2 hegemony, 2 peace, 2 Switzerland	Armed neutrality versus cooperation

```
--seed 0 --output results
uv run analysis.py plot \
--input results/data \
--output results/figures
```

The implementation uses a dedicated pseudorandom generator per run. The automated self-check suite covers growth, combat, territorial transfer, diplomacy, alliance range, unions, reproducibility, collection, and plotting.

4 Results

4.1 Spatial outcomes and dynamics

Figure 1 shows representative final maps for four contrasting scenarios. The outcome is spatially path-dependent even with identical parameter values: an early border contact can create a trade relationship, a retaliatory cycle, or a conquest sequence. Environmental variants visibly change how much land remains available, while composition variants change the number and size of surviving countries.

The baseline time series in Figure 2 clarifies the feedback between population and territory. Initial expansion increases carrying capacity and population. Once borders meet, attacks and pacts alter the balance, and eliminated countries disappear from the trajectories. Territory changes discretely, while population responds gradually through growth, famine, assimilation, and ca-

Table 3: Baseline outcomes over 100 seeds. Territory is the mean number of cells at the end of a run.

Strategy	Survive	Dominant	Territory
HEGEMONY	85%	39%	95.1
RETALIATION	58%	29%	69.3
RANDOM	31%	25%	39.3
PEACE	40%	7%	34.9
SWITZERLAND	47%	0%	16.4

sualties.

4.2 Baseline comparison

Hegemony performs best on all three baseline outcome measures (Table 3 and Figure 3). It survives 85% of runs, is dominant in 39%, and controls 95.1 cells on average. Retaliation is second: 58% survival, 29% dominance, and 69.3 cells. Random behaviour is occasionally dominant but unstable; its median final population is zero. Peace survives in 40% of runs and dominates only 7%. Switzerland avoids offensive casualties but is never dominant in the baseline and controls only 16.4 cells on average. Thus, unconditional non-aggression does not protect a country when an expansionist neighbour is present.

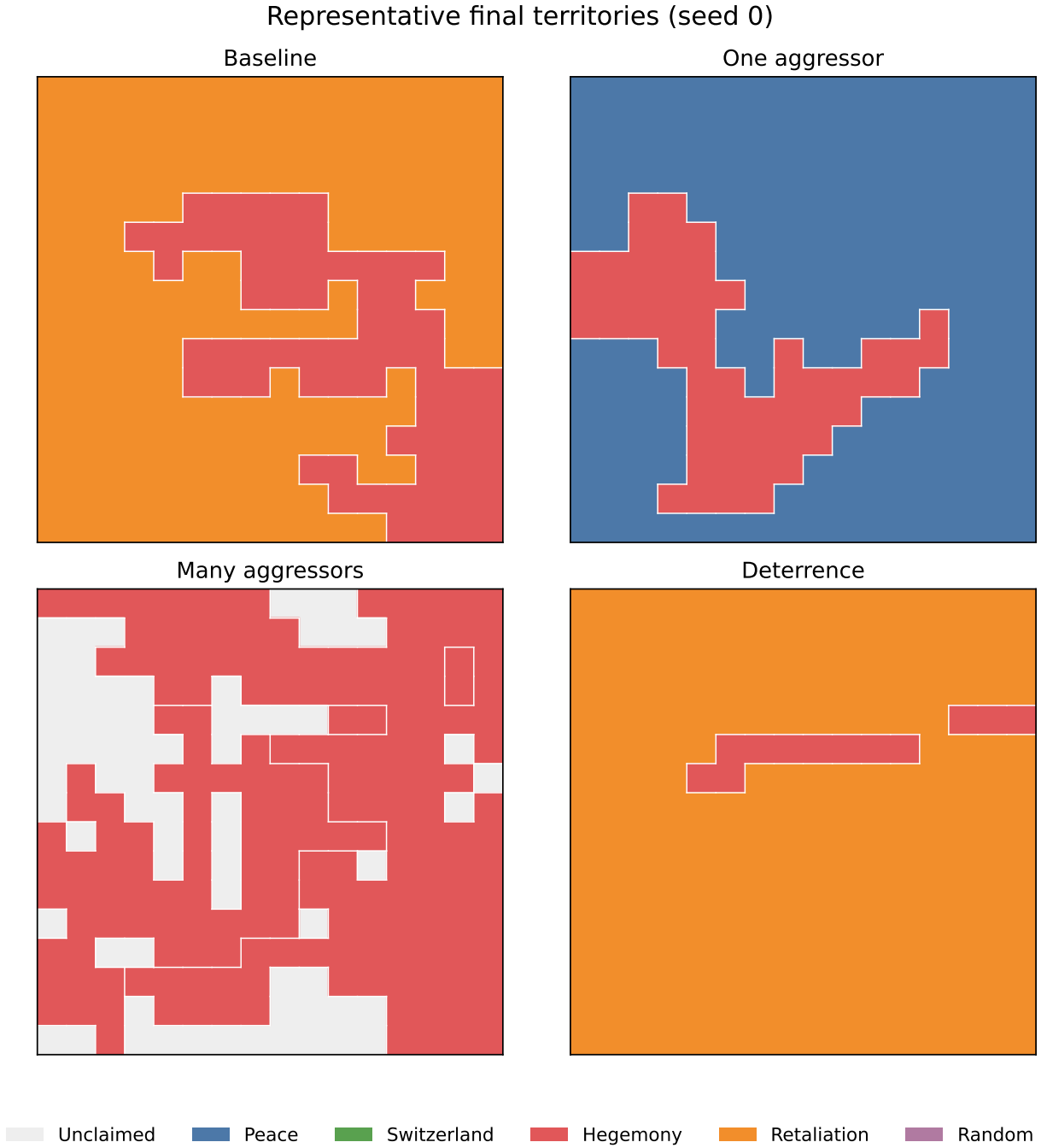


Figure 1: Representative final territories for seed 0. Colours identify strategies; white cells are unclaimed. These maps illustrate possible outcomes, whereas quantitative conclusions use all 100 seeds.

4.3 Environmental conditions

Figure 4 compares the six common-composition environments. Crowding is destructive for every strategy. On the 8×8 map, hegemony is dominant in 45% of runs, while the other strategies have only 5–24% dominance. With abundant space, however, survival becomes common: Switzerland reaches 99%, hegemony 92%, random 73%, and retaliation 68%. Retaliation is dominant in 49% of spacious runs, compared with only 2% for hegemony. More land delays contact and permits populations to grow before conflict; the retaliator then has both resources and a credible response.

Defence changes which strategies can convert survival into dominance. At $d = 3$, hegemony still survives in

99% of runs, but its dominance falls to 28%; peace and retaliation each reach 34% dominance. Switzerland survives 87% but dominates only 1%, showing that survival and expansion are distinct objectives. Random behaviour collapses to 9% survival because indiscriminate attacks become expensive. With no defender advantage ($d = 1$), hegemony controls 104.0 cells on average and dominates 42% of runs.

The continuous sweeps in Figure 5 confirm these patterns using territory share, which remains comparable as the map changes size. From map width 8 to 26, hegemony falls from 44.9% to 19.7% of the map, while retaliation rises from 20.9% to 22.5% and Switzerland from 9.2% to 20.5%. Increasing defence from 1 to 3 reduces hegemony from 40.6% to 33.4%, while peace rises from

Representative baseline dynamics (seed 0)

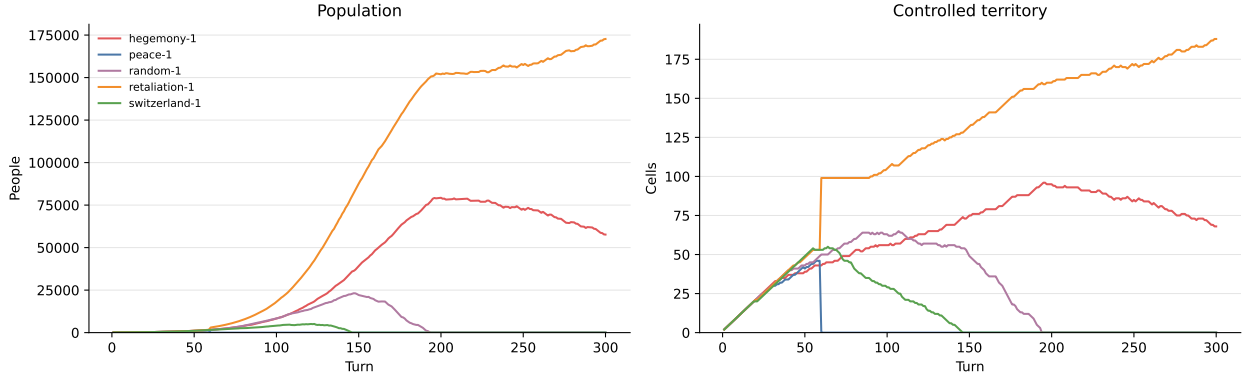


Figure 2: Representative country population and controlled-territory dynamics in the baseline run (seed 0). The figure is descriptive of one trajectory.

Baseline outcomes across repeated runs

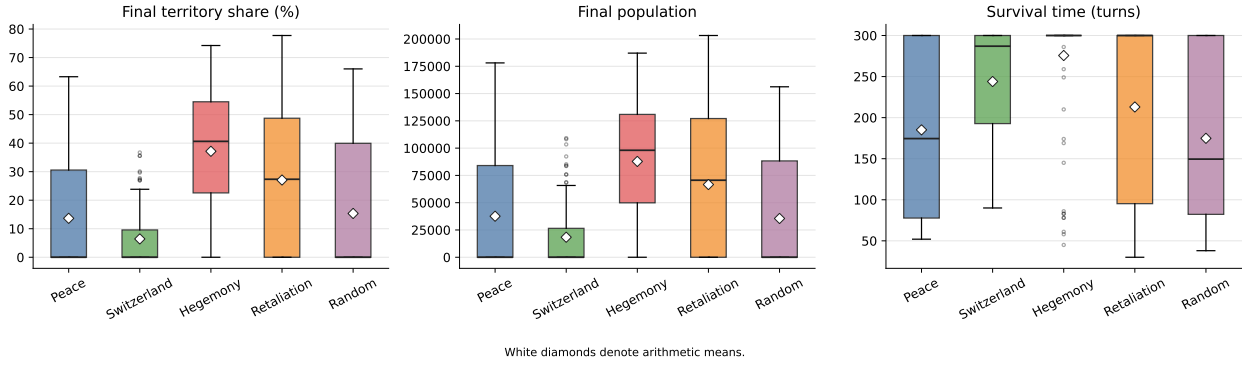


Figure 3: Distributions of final territory, final population, and survival time in the baseline scenario. Each box summarizes 100 country outcomes; white diamonds mark arithmetic means.

Strategy performance across environmental conditions

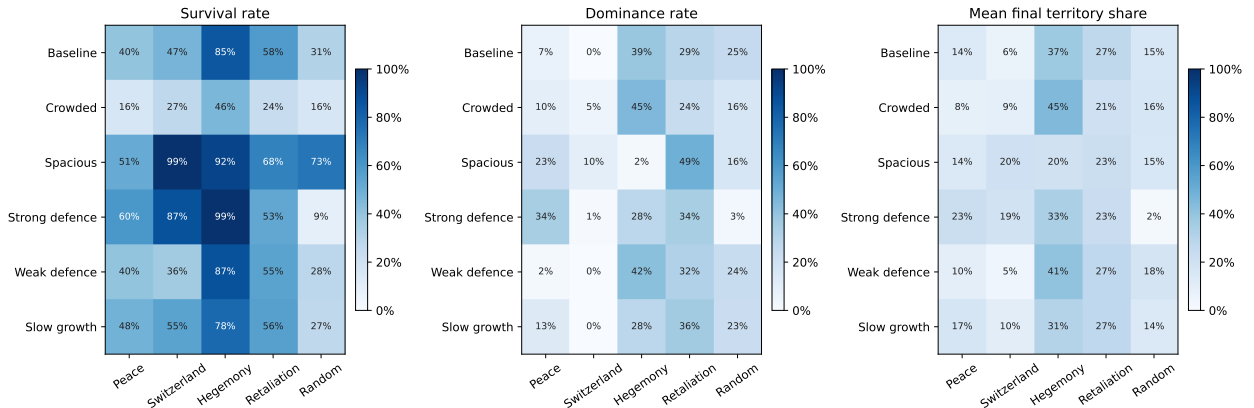


Figure 4: Mean survival, dominance, and final territory share by strategy across six environmental conditions. Each cell summarizes 100 runs.

9.7% to 22.6% and Switzerland from 4.5% to 18.8%. Faster growth benefits hegemony (30.8% at $r = 0.02$ versus 40.4% at $r = 0.10$) but slightly harms peace and Switzerland, because it replenishes the populations used to project military force as well as peaceful populations.

4.4 Strategic context

Performance changes sharply when the mix of neighbours changes (Figure 6). A single hegemony among four peaceful countries survives every run, but the four peaceful countries collectively account for almost all dominance: an individual peace country is dominant in 24.7% of its 400 observations, whereas the hegemony

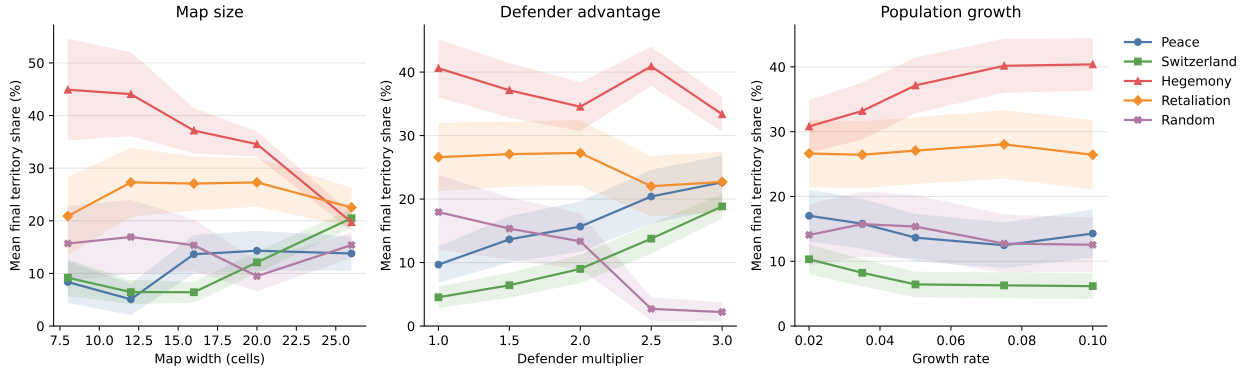


Figure 5: Mean final territory share and 95% Monte Carlo intervals under map-width, defence-multiplier, and population-growth sweeps. Only the parameter on the horizontal axis changes within each panel.

is dominant in only 1% of runs. The aggressor attacks early, but cooperation and occasional unions allow a peaceful country to become the largest successor.

When four hegemonies compete, only 44% survive and each makes 158 attacks on average. The single peace country survives only 19%. In the deterrence scenario, retaliation is dominant in 31% of observations and controls 64.7 cells on average, compared with 12.5% and 38.5 cells for hegemony. Retaliation therefore changes the payoff of aggression: it does not eliminate attacks, but it prevents aggressors from converting them into the same territorial advantage. In the neutral comparison, hegemony remains strongest; peace survives slightly more often than Switzerland (41.5% versus 38.0%), partly because peace can enter military alliances while Switzerland refuses them.

5 Discussion

5.1 What does a group strategy mainly depend on?

The experiments support four answers. First, strategy depends on *resource pressure*. On a crowded map, early contact makes conquest and elimination frequent, strongly favouring hegemony. On a spacious map, countries can grow before meeting, and retaliation becomes more effective than unconditional aggression.

Second, strategy depends on the *relative price of attack*. A high defence multiplier protects peaceful and neutral countries and punishes the random strategy’s unnecessary wars. This is not equivalent to removing conflict: hegemony still survives, but it is less often the largest country.

Third, strategy depends on *population recovery*. Fast growth supplies people for both production and combat. In this model, aggressive countries convert the additional population into territorial gains more effectively, whereas slow growth makes casualties persistent.

Fourth, and most strongly, strategy depends on *who the neighbours are*. Hegemony exploits unconditional pacifism, yet multiple hegemonies impose large costs on each other. Retaliation is weak enough to avoid ini-

tiating war but strong enough to alter an aggressor’s expected gain. Peace can outperform armed neutrality when alliances and unions are valuable. Consequently, “best strategy” is not a property of a strategy alone; it is a conditional property of the spatial, military, demographic, and strategic environment.

These findings should be interpreted mechanistically rather than normatively. Hegemony’s strong baseline result follows from the chosen objective measures and rules: territory raises carrying capacity, and conquest assimilates half of a captured cell’s population. A welfare measure including total deaths, inequality, or pact stability would rank strategies differently. The model therefore demonstrates how assumptions and outcome definitions shape a strategic conclusion.

6 Limitations

The model intentionally favours clarity over realism. Its main limitations are the following.

- The assignment proposes providers, warriors, and scientists whose roles can change. The implementation instead aggregates all members into one population. Technology, labour allocation, and internal adaptation are not modelled; adding them is the most important extension.
- Every cell has equal productivity. The proposed non-uniform resource landscape, food stocks, transfers during scarcity, terrain, travel costs, and capitals are absent. Trade is represented only as a carrying-capacity bonus.
- Countries expand through adjacent cells, but the model does not enforce that a defender’s remaining cells stay connected after conquest. Fragmented territory can therefore occur.
- One country takes one action per turn regardless of population or area. Large states do not have proportionally more administrative or military actions.

Performance depends on neighboring strategies

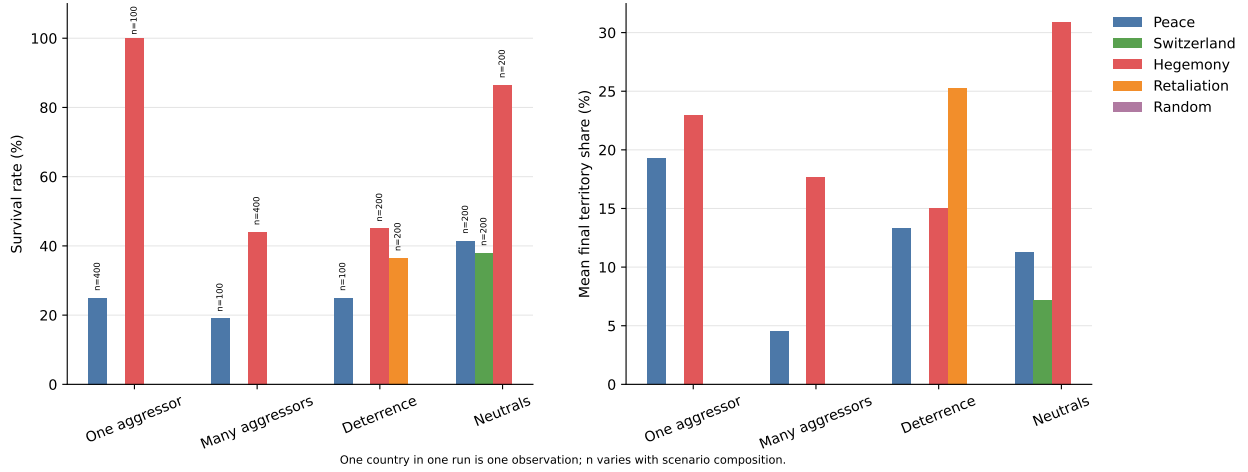


Figure 6: Outcomes under four changes in strategic composition. Labels give the number of observations for each strategy; repeated strategies contribute multiple country observations per seed.

- Strategies are fixed, transparent to the experimenter, and incapable of learning. Countries cannot change doctrine, infer an opponent’s strategy, or negotiate contingent agreements.
- Parameter values are illustrative rather than calibrated to empirical human-group data. Results establish behaviour of this model, not causal claims about real societies.
- Only one-factor sweeps are reported. Interactions between map size, defence, growth, and composition may be nonlinear and require a factorial or global sensitivity analysis.
- Monte Carlo uncertainty reflects seed-to-seed variability only. It does not capture structural uncertainty in the model rules.

7 Conclusions

This project implements a reproducible agent-based simulation of territorial human groups and evaluates five strategies over 2,500 stochastic runs. Aggressive hegemony is the strongest strategy under the baseline assumptions, but its advantage is neither universal nor synonymous with survival. Retaliation becomes the leading strategy when land is abundant or when it directly confronts aggressors. Strong defence makes peaceful cooperation competitive, while neutrality protects survival more reliably than dominance.

The principal conclusion is therefore conditional: a group’s success depends on resource scarcity, the defensive advantage, demographic recovery, and the strategies of neighbouring groups. The separation of data collection from plotting, fixed seeds, run-level outputs, and automated checks make these claims reproducible. Extending the model with explicit providers, warriors, scientists, heterogeneous resources, and adaptive strategies would allow the next study to address the full internal-group dynamics proposed in the original project description.

References

- [1] E. Bonabeau, “Agent-based modeling: Methods and techniques for simulating human systems,” *Proceedings of the National Academy of Sciences*, vol. 99, suppl. 3, pp. 7280–7287, 2002. doi:10.1073/pnas.082080899.
- [2] T. C. Schelling, “Dynamic models of segregation,” *Journal of Mathematical Sociology*, vol. 1, no. 2, pp. 143–186, 1971. doi:10.1080/0022250X.1971.9989794.
- [3] J. M. Epstein and R. L. Axtell, *Growing Artificial Societies: Social Science from the Bottom Up*. Washington, DC: Brookings Institution Press and MIT Press, 1996.
- [4] R. Axelrod and W. D. Hamilton, “The evolution of cooperation,” *Science*, vol. 211, no. 4489, pp. 1390–1396, 1981. doi:10.1126/science.7466396.